

Business Requirements Document (BRD)

Food Ordering System – Web Application

Document Type: Mini Business Requirements Document

Prepared For: Business Analyst Intern Skill Assessment

Project: Food Ordering System

Version: 1.0

1. Project Overview

The Food Ordering System is a web-based platform designed to simplify the process of browsing restaurants and ordering food items online. The system allows customers to register or log in, browse available restaurants and food menus, search for food items, manage their shopping cart, and place orders through an online checkout process.

The system also provides an administrative interface that allows administrators to manage restaurants, food/menu items, item availability, and monitor orders.

The proposed system is intended to provide a fast, user-friendly, and convenient online food ordering experience while giving administrators an efficient way to manage food-related information and orders.

The system will support food categories including **Rice & Curry, Fast Food, Beverages, and Desserts.**

2. Business Objectives

The main objectives of the Food Ordering System are:

1. To simplify the online food ordering process for customers.
2. To allow customers to easily browse restaurants and available food items.
3. To provide customers with an efficient shopping cart and checkout experience.
4. To dynamically calculate the total price of selected food items.
5. To provide restaurants/administrators with a platform to manage food and order-related information.
6. To provide a fast and user-friendly ordering experience.
7. To allow administrators to maintain food item availability and monitor orders.

These objectives are aligned with the purpose stated in the assessment task.

3. Stakeholders and User Roles

3.1 Customer

The customer is the primary end user of the system.

Customers should be able to:

- Register for an account.
- Log in using email and password.
- Optionally use Google social login.
- Browse available restaurants.
- View restaurant food menus.
- Search for food items.
- Add food items to the shopping cart.
- Update item quantities.

- Remove items from the cart.
- View the dynamically calculated total price.
- Place food orders.
- View an order summary.
- Receive order confirmation.

3.2 Administrator

The administrator is responsible for managing the information and operations within the system.

Administrators should be able to:

- Manage restaurants.
- Manage food/menu items.
- Update food item availability.
- Monitor orders.

The assessment specifically identifies Customer and Admin as the two system user roles.

4. Project Scope

4.1 In Scope

The following functions are included within the scope of the Food Ordering System:

- Customer registration and login.
- Optional Google social login.
- Restaurant browsing.
- Food menu viewing.
- Food item searching.
- Shopping cart management.
- Adding items to the cart.
- Updating item quantities.
- Removing items from the cart.
- Dynamic total price calculation.
- Order placement.
- Order summary.
- Order confirmation.
- Restaurant management by administrators.
- Menu/food item management by administrators.
- Food availability management.
- Order monitoring by administrators.

4.2 Out of Scope

The assessment document does not specify detailed functionality for the following areas. Therefore, they are considered outside the defined scope unless additional requirements are provided:

- Online payment gateway integration.
- Delivery partner management.
- Customer reviews and ratings.
- Restaurant delivery tracking.
- Discount and coupon management.

- Loyalty/reward programs.
- Advanced analytics and reporting.

5. Functional Requirements

Functional requirements describe what the system must allow users to do.

ID	Functional Requirement
FR-01	The system shall allow users to register and log in.
FR-02	The system shall allow users to browse available restaurants.
FR-03	The system shall allow users to view restaurant menu items.
FR-04	The system shall allow users to add food items to the shopping cart.
FR-05	The system shall allow users to update or remove cart items.
FR-06	The system shall calculate the total price dynamically based on selected cart items and quantities.
FR-07	The system shall allow users to place food orders.
FR-08	The system shall allow administrators to manage food items.
FR-09	The system shall allow administrators to manage restaurants.
FR-10	The system shall allow administrators to update food item availability.
FR-11	The system shall allow administrators to monitor orders.
FR-12	The system shall provide an order summary before/after order placement.

FR-01 to FR-08 are based on the functional requirements explicitly provided in the assessment, while FR-09 to FR-12 elaborate the stated Admin features and system flow.

6. Non-Functional Requirements

Non-functional requirements define the expected quality and operational characteristics of the system.

6.1 Security

The system should provide secure authentication for customer and administrator accounts. User credentials should be protected from unauthorized access.

6.2 Performance

The system should provide fast loading and responsive interactions when customers browse restaurants, view menus, manage carts, and proceed through the ordering process.

6.3 Usability

The system should have a simple and user-friendly interface. The web application should support a mobile-responsive design so that customers can access the system using different screen sizes.

6.4 Reliability

The system should provide proper error handling when users perform actions such as logging in, adding items to the cart, updating quantities, or placing orders.

These four non-functional areas—Security, Performance, Usability, and Reliability—are explicitly identified in the assessment.

7. Business Rules

The following business rules are derived from the stated system features and flow:

BR-01: Customers must be authenticated before completing the checkout/order placement process.

BR-02: Customers can add available food items to their shopping cart.

BR-03: Customers can modify the quantity of items in their cart.

BR-04: Customers can remove unwanted items from their cart.

BR-05: The cart total should change dynamically when item quantities or selected items are changed.

BR-06: Only available food items should be selectable for ordering.

BR-07: An administrator is responsible for maintaining restaurant and food/menu information.

BR-08: Administrators can update the availability status of food items.

BR-09: An order should contain the selected food items and the applicable total shown during checkout.

BR-10: Customers should receive an order confirmation after successfully placing an order.

The assessment's stated flow is: **Browse Restaurants → Select Food → Add to Cart → Login/Register → Checkout → Order Confirmation.**

8. High-Level User Flow

The high-level customer journey is:

Browse Restaurants

↓

Select Restaurant

↓

View Food Menu

↓

Search/Select Food Item

↓

Add Item to Cart

↓

Manage Cart / Update Quantities

↓

Login/Register

↓

Checkout

↓

Place Order

↓

Order Confirmation

The above flow follows the simple system flow supplied in the assessment.

9. Assumptions

- Customers will access the system through a web browser.
- Customers may browse restaurants and add food items to their cart before authentication, as indicated by the provided system flow.
- Restaurants and food items will be maintained by authorized administrators.
- Customers will provide valid registration information.
- Food item availability will be maintained by the administrator.
- The system will calculate order totals based on items and quantities selected by the customer.
- Google login is considered an optional authentication method.

10. Acceptance Criteria

The solution should be considered to meet the core business requirements when:

1. A customer can successfully register and log in.
2. A customer can browse restaurants and view available menus.
3. A customer can search for food items.
4. A customer can add food items to a cart.
5. A customer can update quantities and remove items.
6. The system dynamically updates the cart total.
7. A customer can proceed through checkout and place an order.
8. An order summary and confirmation are provided after successful ordering.
9. An administrator can manage restaurants and menu items.
10. An administrator can update food availability and monitor orders.
11. The system provides secure authentication, acceptable performance, mobile responsiveness, and proper error handling.

11. Conclusion

The proposed Food Ordering System provides a centralized web-based solution for customers to browse restaurants, select food items, manage their shopping carts, and place food orders online. At the administrative level, the system provides functionality for managing restaurants, food items, availability, and orders.

The requirements analysis establishes the key business needs, user roles, functional requirements, non-functional requirements, business rules, and acceptance criteria for the proposed system. These requirements will provide the foundation for the system diagram, Work Breakdown Structure, five-day project timeline, and final project summary.

The overall solution aims to simplify the food ordering process while providing a fast, user-friendly, secure, and reliable web experience.